

PORT ALFRED, South Africa

WMO: 688430

Lat: 33.559S

Lon: 26.881E

Elev: 276

StdP: 14.55

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	41.5	43.9	29.5	23.8	65.7	32.4	27.1	61.0	24.9	61.7	21.9	59.2	6.1	280	0.396

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.1	85.3	67.1	81.9	67.7	79.2	67.7	73.1	80.1	71.7	78.4	70.6	76.9	12.2	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.9	115.0	76.3	69.5	109.6	75.6	68.2	104.5	74.7	36.9	80.0	35.7	78.5	34.7	77.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.5	20.7	18.2	38.0	94.0	3.2	3.7	35.7	96.6	33.8	98.8	32.0	100.8	29.7	103.5		
(5)			35.1	75.2	2.0	0.8	33.6	75.7	32.5	76.1	31.3	76.6	29.9	77.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	63.9	70.2	70.7	68.8	64.8	62.1	58.8	58.5	60.1	62.5	64.5	67.8	(6)	
(7)		DBStd	6.09	3.49	3.85	4.08	4.49	4.36	4.84	5.37	4.55	3.89	4.13	3.67	3.73	(7)
(8)		HDD50	4	0	0	0	0	1	1	1	1	0	0	0	0	(8)
(9)		HDD65	1138	3	3	11	54	114	199	219	210	156	101	53	15	(9)
(10)		CDD50	5072	626	579	582	445	374	264	263	264	302	386	434	553	(10)
(11)		CDD65	732	164	162	128	49	22	13	17	9	7	22	37	102	(11)
(12)		CDH74	2658	559	568	382	205	141	116	137	76	49	88	65	272	(12)
(13)		CDH80	511	76	98	81	56	34	27	32	26	13	19	6	43	(13)
(14)	Wind	WSAvg	8.1	9.0	8.5	7.5	7.0	6.8	7.8	7.6	7.7	8.2	9.0	9.0	8.9	(14)
(15)	Precipitation	PrecAvg	24.9	1.6	2.4	2.5	2.1	1.5	2.1	1.7	2.2	3.0	2.4	1.7	(15)	
(16)		PrecMax	37.3	2.5	5.1	5.0	5.9	5.0	7.4	5.8	11.1	7.5	6.7	6.3	5.0	(16)
(17)		PrecMin	18.1	0.8	0.9	0.5	0.2	0.3	0.2	0.1	0.6	0.4	0.9	0.7	0.3	(17)
(18)		PrecStd	5.9	0.6	1.1	1.3	1.7	1.3	1.8	1.8	2.7	2.0	1.7	1.5	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.4	88.6	88.1	86.6	84.4	84.1	84.1	83.9	80.3	82.0	80.2	85.2	(19)
(20)			MCWB	72.4	72.0	71.3	67.4	61.0	57.8	57.6	57.6	61.7	65.4	67.3	70.2	(20)
(21)		2%	DB	81.6	82.5	81.4	79.1	77.4	77.1	77.7	74.1	72.9	75.1	75.6	79.5	(21)
(22)			MCWB	71.4	71.4	69.7	66.2	61.9	56.0	57.4	58.1	62.2	64.6	65.9	69.1	(22)
(23)		5%	DB	79.0	79.3	77.7	74.9	72.8	71.3	71.2	69.6	69.4	71.4	73.4	76.5	(23)
(24)			MCWB	70.3	70.5	68.7	65.3	60.7	55.4	56.0	58.3	61.4	63.3	65.1	67.7	(24)
(25)		10%	DB	76.8	77.2	75.3	71.8	69.4	67.3	67.1	66.5	67.0	69.3	71.6	74.5	(25)
(26)			MCWB	68.8	69.6	68.0	64.3	60.5	56.4	55.8	57.7	59.9	62.1	64.3	66.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.9	74.5	74.5	70.9	67.1	63.6	64.1	64.2	66.3	68.8	70.1	72.1	(27)
(28)			MCD B	80.1	81.8	82.2	78.4	75.6	67.5	70.6	71.1	72.5	78.4	77.1	80.9	(28)
(29)		2%	WB	72.7	72.9	72.3	68.7	64.8	61.0	61.0	62.0	64.3	66.4	68.2	70.2	(29)
(30)			MCD B	79.7	79.8	78.2	75.3	71.5	68.0	68.1	68.2	70.0	72.4	74.0	77.4	(30)
(31)		5%	WB	71.4	71.6	70.4	67.1	63.2	59.0	59.1	60.4	62.6	64.7	66.4	68.8	(31)
(32)			MCD B	77.9	78.2	75.9	73.3	69.2	67.3	67.3	66.1	67.9	69.9	72.0	75.1	(32)
(33)		10%	WB	70.0	70.4	69.0	65.5	61.8	57.4	57.6	58.9	60.9	63.1	64.9	67.5	(33)
(34)			MCD B	76.0	76.7	74.3	71.3	67.7	65.6	64.9	64.9	66.1	67.9	70.5	73.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.6	12.1	12.5	14.3	16.8	18.6	18.3	17.0	14.5	12.4	12.4	11.9	(35)
(36)			MCD BR	16.2	15.8	16.8	20.7	23.8	26.0	26.3	23.3	19.3	17.4	15.8	15.5	(36)
(37)			MCWBR	8.8	8.0	8.5	10.4	11.7	11.2	11.7	12.3	11.3	10.3	9.7	8.9	(37)
(38)		5% WB	MCD BR	15.2	14.3	14.6	16.5	18.3	19.4	19.9	17.8	16.9	15.9	14.6	14.8	(38)
(39)			MCWBR	8.7	7.7	8.3	9.6	11.3	11.9	12.2	11.4	11.0	9.8	9.1	8.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.397	0.398	0.383	0.357	0.341	0.326	0.330	0.374	0.397	0.408	0.388	0.391	(40)	
(41)		taud	2.441	2.446	2.471	2.512	2.527	2.539	2.525	2.392	2.324	2.331	2.424	2.446	(41)	
(42)		Ebn at Noon	298	292	284	274	262	258	263	264	275	285	299	301	(42)	
(43)		Edh at Noon	38	37	34	29	25	23	25	32	38	41	39	38	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2112	1932	1611	1249	969	867	924	1160	1503	1770	2042	2184	(44)	
(45)		RadStd	108	99	123	66	41	45	55	72	102	146	116	136	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		+0.63	+0.76	N/A	N/A	N/A	N/A	N/A	-153	+256	+81

Nomenclature: See separate page